



AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2024 – 0423



Agricultural Machinery – Soybean Roaster DAVAO TECHNOCRAFT GM-SBR2024

Test Report Number	2024 – 0423
Date of Issuance	September 25, 2024
Valid Until	September 25, 2029
Agricultural Machinery	Soybean Roaster
Type	Horizontal drum, Batch type, Electric motor-driven, LPG-fed
Brand	DAVAO TECHNOCRAFT
Model	GM-SBR2024
Test Requested by	DAVAO TECHNO CRAFT 424 Echavez St., P-4 Poblacion, Nabunturan, Davao De Oro

SUMMARY

Performance Criteria	Manufacturer's Specification	AMTEC Test
Input capacity, kg/h	ND ^a	74.51
Output capacity, kg/h	ND ^a	67.76
Roasting recovery, %, minimum	ND ^a	90.80
Mechanically damaged beans, %, minimum	ND ^a	0.543

^a Not provided by the test applicant.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	DAVAO TECHNO CRAFT
4. Serial number	SBRELEC-05
5. Prime mover used	0.373 kW MEGA YC 80B-4 Electric Motor
6. Number of operators	Two (One feeding operator/machine controller and one feeding assistant/bagger)
7. Loading of input	The soybeans were manually loaded into the input hopper using steel bins.
8. Cleaning of parts	Parts are accessible for cleaning. Some components and covers needs to be removed to cleaned properly.
9. Adjusting and repairing of parts	a. Locally fabricated parts can be repaired and replaced. b. Heat can be adjusted through the burner's valves. c. Rotation of the roasting drum can be reversed for discharge process of roasted beans. d. Belt tension can be adjusted by moving the prime mover along its base.
10. Collecting output	The output from the soybean roaster was cooled in the cooling tray (Figure 3) and then collected using steel bins.
11. Transporting the machine	The machine is stationary. It can be transported by manual lifting.
12. Safety features	a. Transmission system cover is provided. b. On/Off and reverse rotation buttons. c. Shut off valve for the heat source.

OBSERVATIONS (Cont'n)

13. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	No failure or abnormalities were observed on the machine or its components during and after the operation.
14. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Date of Test	September 4, 2024
Location of Test	Prk.4, Brgy. Poblacion, Nabunturan, Davao de Oro 7° 36' 20.22" N, 125° 57' 55.52" E
Items Tested	Operator's Manual, Input Capacity, Output Capacity, Roasting Recovery, Roasting Temperature, Power Consumption, Specific Energy Consumption, Fuel Consumption, Speed of Components, Noise Level, Moisture Content Variation, Moisture Content Reduction Rate, Mechanically Damaged Beans, and Number of Operators

METHODS OF TEST

The coffee roaster was tested based on the PNS/BAFS 215:2023 – Agricultural Machinery – Green Coffee Bean Roaster – Methods of Test.

DESCRIPTION OF THE MACHINE

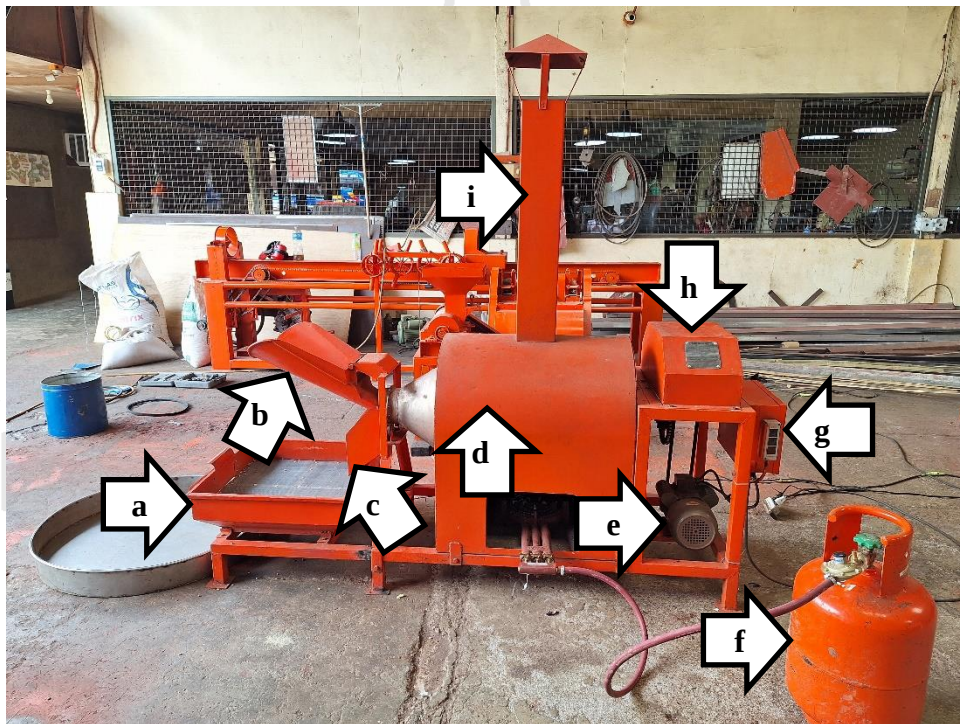


Figure 1. Parts of the DAVAO TECHNOCRAFT GM-SBR2024 Soybean Roaster powered by 0.373 kW MEGA YC 80B-4 Electric Motor include the following: (a) Cooling tray, (b) Input hopper, (c) Output chute, (d) Roasting chamber, (e) Prime mover, (f) Heat source, (g) Control buttons, (h) Transmission system cover, and (i) Chimney.

SPECIFICATIONS

Item		Manufacturer's Specifications ^a	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	1800	2100
1.1.2	Width	660	920
1.1.3	Height	1840	1815
1.2	Weight, kg	ND	NM ^b
2	Input capacity, kg/h	ND	74.51
3	Heat source	LPG – Fire single burner	LPG - Liquefied Petroleum Gas
4	Input hopper		
4.1	Overall dimension, mm		
4.1.1	Top opening dimensions, L × W, mm	525 × 315	350 × 310
4.1.2	Bottom opening dimension, L × W, mm	ND	97.72 × 57.86
4.1.3	Height	75	85
4.2	Height from the ground, mm	765	770
4.3	Material	G.I plain sheet (1.3 mm)	Iron/Carbon Steel ^c (t = 1.42 mm)
4.4	Location	Front	Side of the roasting chamber
4.5	Means of attachment	All part bolted	Bolted
5	Output chute		
5.1	Overall dimension, mm		
5.1.1	Length	200	165
5.1.2	Width	155	170
5.1.3	Height	ND	170
5.2	Height from the ground, mm	258	345
5.3	Material	G.I plain sheet (1.3 mm)	Iron/Carbon Steel ^c (t = 1.11 mm)
5.4	Location	Front	Side of the roasting chamber
5.5	Means of attachment	Welded	Bolted

^a The data were obtained from the specifications sheet provided by the test applicant.

^b No available test instrument to measure the required parameter.

^c Determined using the BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for elemental composition.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specifications ^a	AMTEC Verification
6	Roasting mechanism		
6.1	Type	Horizontal rotary drum roaster (direct fire)	Horizontal drum roaster
6.2	Dimension, L × D × t, mm	ND × 515 × ND	765 × 514 × 1.41
6.3	Material		
6.3.1	Roasting drum	Stainless steel #304 (1.2 mm)	Nitronic32 ^b (t = 1.41 mm)
6.3.2	Roasting drum cover	ND	Iron/Carbon Steel ^b
7	Cooling table		
7.1	Dimension, L × W × H, mm	ND	525 × 525 × 85
7.2	Material	ND	Iron/Carbon Steel ^b (t = 1.11 mm)
7.3	Capacity, kg	25	22.00 ^c
8	Prime mover		
8.1	Brand	MEGA	MEGA
8.2	Model	YC 80B-4	YC 80B-4
8.3	Make	China	China
8.4	Rated power, kW	0.373	0.373 ^a
8.5	Rated speed, rpm	1740	1740 ^a
8.6	Electric service required	Single-phase	Single-phase
8.7	Line voltage, V	220	220 ^a
8.8	Line current, A	4.5	4.5 ^a
8.9	Frequency, Hz	60	60 ^a
9	Safety device	Push button forward & reverse switch & magnetic contactor	a. Transmission system cover b. On/Off buttons c. Shut off valve for the heat source.
10	Special features	Had forward reversed loading and unloading of roasted soybeans	a. Blower for the cooling tray is provided. b. Roasting drum reverse rotation button

^a The data were obtained from the motor nameplate and specifications sheet provided by the test applicant.

^b Determined using the BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for elemental composition.

^c Weight of output during actual test.

Note: ND – No Data

RESULTS

Item		Data	
1	Crop condition		
1.1	Type	Soybeans	
1.2	Source	Tago, Surigao del Sur	
1.3	Variety	ND ^a	
1.4	Moisture content, % _{w.b.}	11.15	
1.5	Dimensions, L × W × t, mm	8.45 × 4.92 × 5.97	
1.6	Bulk density, kg/m ³	714.96	
1.7	Purity, %	99.25	
2	Performance test		
2.1	Weight of input, kg	25.00	
2.2	Weight of output, kg	22.00	
2.3	Roasting time, min	20.19	
2.4	Input capacity, kg/h	74.51	
2.5	Output capacity, kg/h	67.76	
2.6	Roasting recovery, %	90.80	
2.7	Specific energy consumption, kJ/kg	3510.64	
2.8	Speed of components, rpm	Without load	With load
2.8.1	Prime mover	1786	1772
2.8.2	Roasting Chamber	30	29
2.8.3	Speed reducer-Input	1785	1769
2.8.4	Speed reducer-Output	88	88
2.9	Noise level, dB(A)	Without load	With load
2.9.1	Feeding operator's ear level	69.4	68.4
2.9.2	Bagger's ear level	69.7	71.8
2.10	Power consumption	Without load	With load
2.10.1	Power, kW	0.466	0.526
2.10.2	Voltage, V	229.9	223.7
2.10.3	Current, A	2.03	2.35
2.11	LPG fuel consumption, kg/h	5.34	
2.12	Moisture content reduction rate, %/h	7.98	
3	Laboratory analysis of output		
3.1	Output dimensions, L × W × t, mm	8.43 × 5.06 × 6.21	
3.2	Output bulk density, kg/m ³	714.67	
3.3	Moisture content, % _{w.b.}	8.49	
3.4	Moisture content variation at the end of roasting, %	2.18	
3.5	Mechanically damaged beans, %	0.543	

^a Data were not provided by the test applicant.

Note: ND – No Data

RESULTS (Cont'n)

Table 1. Elemental composition of the different components of the DAVAO TECHNOCRAFT GM-SBR2024 Soybean Roaster.

Element	Percent Composition ^a						
	Input hopper	Cooling tray	Output chute	Roasting drum cover	Roasting drum	Chimney	Main frame
Vanadium	0.015	0.017	0.016	0.014	0.125	0.015	0.017
Chromium	<LOD	<LOD	<LOD	<LOD	14.638	<LOD	0.027
Manganese	0.281	0.133	0.285	0.290	10.680	0.283	0.353
Iron	99.664	99.781	99.632	99.542	72.572	99.636	99.353
Cobalt	<LOD	<LOD	0.060	0.143	<LOD	<LOD	0.177
Nickel	<LOD	<LOD	<LOD	<LOD	1.086	<LOD	0.014
Copper	0.018	0.020	0.016	0.019	0.838	0.020	0.025
Zirconium	-	-	-	-	<LOD	-	-
Niobium	<LOD	<LOD	<LOD	<LOD	0.004	<LOD	<LOD
Molybdenum	0.011	0.012	0.011	0.012	0.016	0.012	0.011
Tin	<LOD	<LOD	<LOD	<LOD	<LOD	0.016	<LOD
Tantalum	-	-	-	-	<LOD	-	-
Tungsten	0.012	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD
Lead	-	0.047	0.031	0.036	0.014	0.032	0.033
Titanium	-	-	-	-	0.024	-	-
Grade	Iron/ Carbon Steel	Iron/ Carbon Steel	Iron/ Carbon Steel	Iron/ Carbon Steel	Nitronic32	Iron/ Carbon Steel	1000 Series LAS

^a Determined using the BRUKER S1 TITAN 500 X-Ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

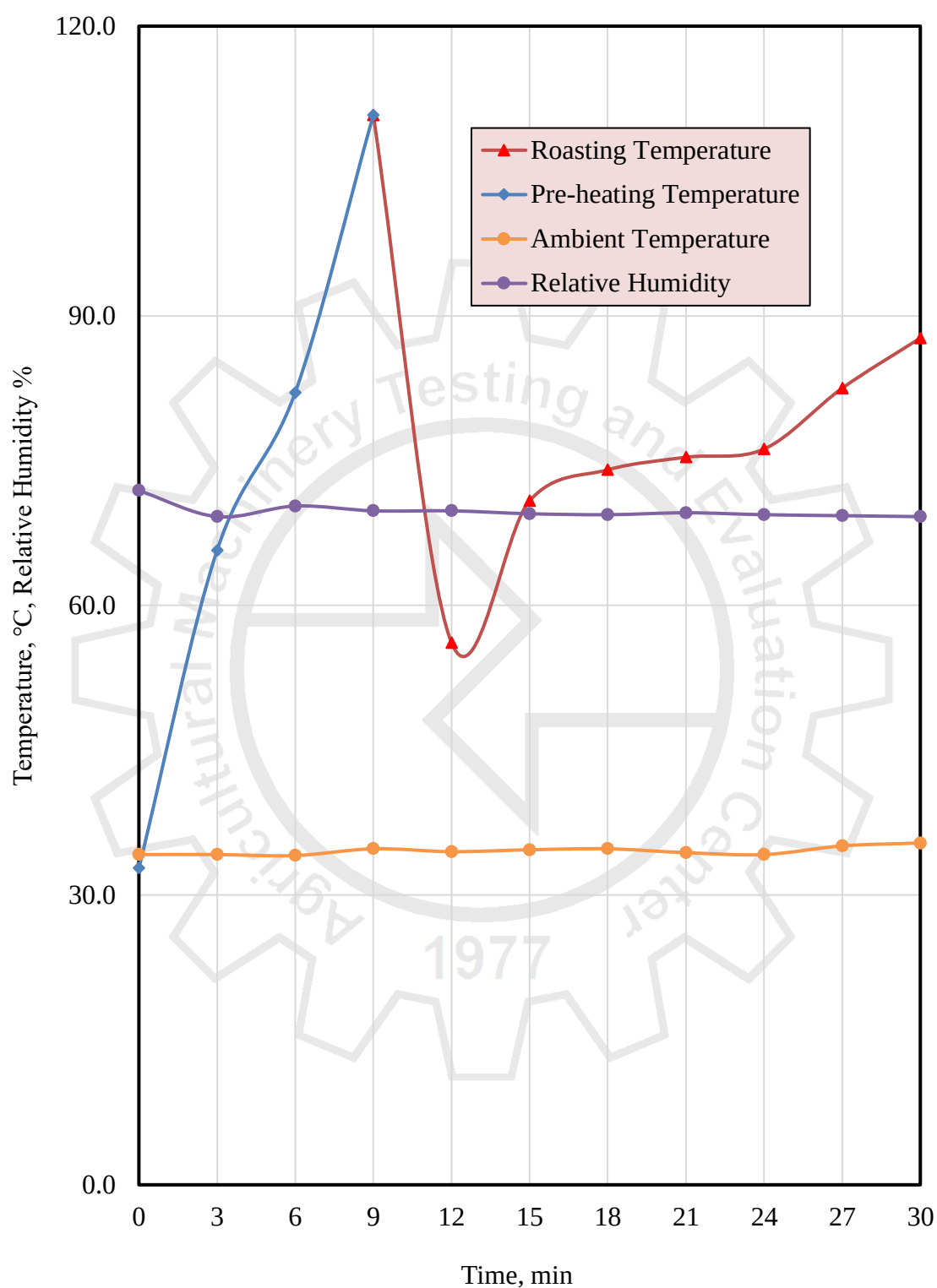
RESULTS (Cont'n)

Figure 2. Roasting temperature curve for the DAVAO TECHNOCRAFT GM-SBR2024 Soybean Roaster.

OBSERVATIONS



Figure 3. Quality of output.



Figure 4. Nameplate of the DAVAO TECHNOCRAFT GM-SBR2024 Soybean Roaster.

OBSERVATIONS (Cont'n)

Figure 5. Labels and safety warning stickers on the DAVAO TECHNOCRAFT GM-SBR2024 Soybean Roaster.

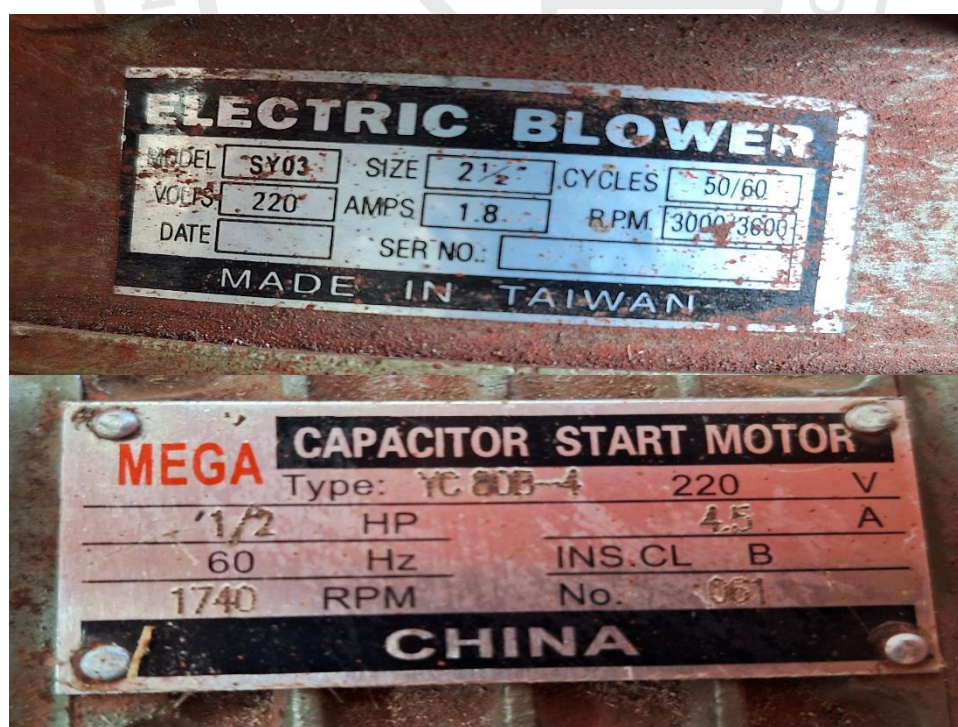


Figure 6. Nameplate of the blower (top) and prime mover (bottom) of the DAVAO TECHNOCRAFT GM-SBR2024 Soybean Roaster.

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02 OCT 2024

Date

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